Free fermions $\hat{H}_0 = \sum_{\vec{k}} \left(\frac{\hbar^2 \vec{k}^2}{2m} - \mu \right) \underbrace{\hat{\psi}^+_k \hat{\psi}_k}_{\text{operators}}$

$$\Rightarrow \quad Z = \int D\bar{\psi} D\psi \; \exp \sum_{\vec{k}, \omega_n} \left(i\omega_n - \frac{\hbar^2 \vec{k}^2}{2m} + \mu \right) \underbrace{\bar{\psi}_k(\omega_n) \psi_k(\omega_n)}_{\text{Grassmann numbers}}$$

Interacting fermions $\hat{H}_0 \to \hat{H}_0 + \hat{H}_{int}$

<u>Superconductivity</u> (reduced BCS Hamiltonian)

$$\hat{H}_{int} = -\frac{g}{V} \sum_{\vec{k}, \vec{k}'} \hat{\psi}^+_{k\uparrow} \hat{\psi}^+_{-k\downarrow} \hat{\psi}_{-k'\downarrow} \hat{\psi}_{k'\uparrow}$$

spin indices $\uparrow, \downarrow$

In the action we have $\int_0^\beta d\tau \, H = -\frac{g}{V} \sum_{kk'} \int_0^\beta d\tau \, \hat{\psi}^+_{k\uparrow} \hat{\psi}^+_{-k\downarrow} \hat{\psi}_{-k'\downarrow} \hat{\psi}_{k'\uparrow}$

$\Rightarrow$ Fourier transform in τ:

$$\psi_{k\sigma}(\tau) = \frac{1}{\sqrt{\beta}} \sum_{\omega} e^{-i\omega\tau} \psi_{k\sigma}(\omega)$$

$$\Rightarrow -\frac{g}{V} \sum_{kk'} \frac{1}{\beta^2} \sum_{\substack{\omega_1 \omega_2 \\ \omega_3 \omega_4}} \int_0^\beta d\tau \; e^{i(\omega_1 + \omega_2 - \omega_3 - \omega_4)\tau} \hat{\psi}^+_{k\uparrow}(\omega_1) \hat{\psi}^+_{-k\downarrow}(\omega_2) \hat{\psi}_{-k'\downarrow}(\omega_3) \hat{\psi}_{k'\uparrow}(\omega_4)$$

$$\Rightarrow \quad \omega_1 + \omega_2 = \omega_3 + \omega_4 = \Omega$$

$$\omega_1 = \omega + \Omega/2, \quad \omega_2 = -\omega + \Omega/2$$

$$\omega_3 = -\omega' + \Omega'/2, \quad \omega_4 = \omega' + \Omega'/2$$

$$-\frac{g}{\beta V} \sum_{\vec{k}\vec{k}'} \sum_{\omega \Omega \Omega'} \hat{\psi}^+_{k\uparrow}\left(\omega + \frac{\Omega}{2}\right) \hat{\psi}^+_{-k\downarrow}\left(-\omega + \frac{\Omega}{2}\right) \hat{\psi}_{-k'\downarrow}\left(-\omega + \frac{\Omega'}{2}\right) \hat{\psi}_{k'\uparrow}\left(\omega + \frac{\Omega'}{2}\right)$$

Assume local pairing ($g>0$)

$$H_{int} = -g \int d\vec{r}\, \hat{\psi}^+_\uparrow(r)\, \hat{\psi}^+_\downarrow(r)\, \hat{\psi}_\downarrow(r)\, \hat{\psi}_\uparrow(r) =$$

$$= -g \int d\vec{r}\, \frac{1}{V^2} \sum_{k_1 k_2 k_3 k_4} \hat{\psi}^+_{k_1\uparrow}\, \hat{\psi}^+_{k_2\downarrow}\, \hat{\psi}_{k_3\downarrow}\, \hat{\psi}_{k_4\uparrow}\, e^{-i\vec{r}(\vec{k}_1+\vec{k}_2-\vec{k}_3-\vec{k}_4)}$$

$$\Rightarrow \vec{k}_1+\vec{k}_2 = \vec{k}_3+\vec{k}_4$$

let $\vec{k}_1+\vec{k}_2 = \vec{q} = \vec{k}_3+\vec{k}_4$

$$\vec{k}_1 = \vec{k}+\vec{q}/2\,,\quad \vec{k}_2 = -\vec{k}+\vec{q}/2$$

$$\vec{k}_3 = -\vec{k}'+\vec{q}/2\,,\quad \vec{k}_4 = \vec{k}'+\vec{q}/2$$

$$H_{int} = -\frac{g}{V} \sum_{\vec{k}\vec{k}'\vec{q}} \hat{\psi}^+_{k+\frac{q}{2}\uparrow}\, \hat{\psi}^+_{-k+q/2\downarrow}\, \hat{\psi}_{-k'+\frac{q}{2}\downarrow}\, \hat{\psi}_{k'+\frac{q}{2}\uparrow}$$

Also transform in τ:

$$\hat{\psi}_k(\tau) = \frac{1}{\sqrt{\beta}} \sum_\omega e^{-i\omega\tau}\, \hat{\psi}_k(\omega)$$

$$\Rightarrow \int_0^\beta d\tau\, \hat{H}_{int} = -\frac{g}{\beta V} \sum_{\vec{k}\vec{k}'\vec{q}} \sum_{\omega\omega'\Omega} \hat{\psi}^+_{k+\frac{q}{2}\uparrow}\left(\omega+\frac{\Omega}{2}\right) \hat{\psi}^+_{-k+\frac{q}{2}\downarrow}\left(-\omega+\frac{\Omega}{2}\right) \cdot$$

$$\cdot\, \psi_{-k'+\frac{q}{2}\downarrow}\left(-\omega'+\Omega/2\right) \psi_{k'+q/2\uparrow}\left(\omega'+\frac{\Omega}{2}\right)$$

$\Rightarrow$ can introduce 4-vector $(\vec{k},\omega) = k$

$$Z_F = \int D\bar{\psi} D\psi \; \exp \int_0^\beta d\tau \Big\{ \sum_{k\sigma} \bar{\psi}_{k\sigma} \Big(-\frac{\partial}{\partial \tau} - (\varepsilon_k - \mu) \Big) \psi_{k\sigma} +$$

$$+ \frac{g}{V} \sum_{\vec{k}\vec{k}'} \bar{\psi}_{k\uparrow + \frac{q}{2}}(\tau) \, \bar{\psi}_{-k\downarrow + \frac{q}{2}}(\tau) \, \psi_{-k'\downarrow + \frac{q}{2}}(\tau) \, \psi_{k'\uparrow + \frac{q}{2}}(\tau) \Big\}$$

let us introduce

$$Z_B = \int D\bar{\Phi} D\phi \; \exp - \int_0^\beta d\tau \sum_q \frac{1}{\lambda} \bar{\Phi}_q(\tau) \phi_q(\tau)$$

And shift
$$\begin{cases} \phi_q(\tau) \to \phi_q(\tau) + \dfrac{\lambda}{\sqrt{V}} \displaystyle\sum_k \psi_{-k+\frac{q}{2}\downarrow}(\tau) \, \psi_{+k'+\frac{q}{2}\uparrow}(\tau) \\ \bar{\Phi}_q(\tau) \to \bar{\Phi}_q(\tau) + \dfrac{\lambda}{\sqrt{V}} \displaystyle\sum_{k'} \bar{\psi}_{+k+\frac{q}{2}\uparrow}(\tau) \, \bar{\psi}_{k+\frac{q}{2}\downarrow}(\tau) \end{cases}$$

$$\frac{1}{\lambda} \sum_q \bar{\Phi}_q(\tau) \phi_q(\tau) \to \frac{1}{\lambda} \sum_q \Big\{ \bar{\Phi}_q(\tau) \phi_q(\tau) + \frac{\lambda}{\sqrt{V}} \phi_q(\tau) \sum_{k'} \bar{\psi}_{k+\frac{q}{2}\uparrow}(\tau) \bar{\psi}_{-k+\frac{q}{2}\downarrow}(\tau)$$

$$+ \frac{\lambda}{\sqrt{V}} \bar{\phi}_q(\tau) \sum_{k'} \psi_{-k'+\frac{q}{2}\downarrow}(\tau) \, \psi_{k'+\frac{q}{2}\uparrow}(\tau)$$

$$+ \frac{\lambda^2}{V} \sum_{kk'} \bar{\psi}_{k+\frac{q}{2}\uparrow}(\tau) \bar{\psi}_{-k+\frac{q}{2}\downarrow}(\tau) \, \psi_{-k'+\frac{q}{2}\downarrow}(\tau) \, \psi_{k'+\frac{q}{2}\uparrow}(\tau) \Big\}$$

Now choose λ such that to <u>cancel</u> the 4-fermion term in $Z_F Z_B$: $\Rightarrow \underline{\lambda = g}$

$$Z = Z_F Z_B = \int D\bar{\psi} D\psi D\bar{\Phi} D\phi \, .$$

$$\exp \int_0^\beta d\tau \Big\{ \sum_{k\sigma} \bar{\psi}_{k\sigma} \Big(-\frac{\partial}{\partial \tau} - (\varepsilon_k - \mu) \Big) \psi_{k\sigma} - \frac{1}{g} \sum_q \bar{\Phi}_q(\tau) \phi_q(\tau)$$

$$- \frac{1}{\sqrt{V}} \sum_{qk} \bar{\Phi}_q(\tau) \psi_{-k'+\frac{q}{2}\downarrow}(\tau) \psi_{k'+\frac{q}{2}\uparrow}(\tau)$$

$$- \frac{1}{\sqrt{V}} \sum_{qk'} \phi_q(\tau) \bar{\psi}_{k+\frac{q}{2}\uparrow}(\tau) \bar{\psi}_{-k+\frac{q}{2}\downarrow}(\tau) \Big\}$$

F.T. gives: (fermion) $\omega_n = \frac{2\pi}{\beta}\left(n+\frac{1}{2}\right)$; $\omega_m = \frac{2\pi}{\beta} m$ (boson)

$$Z = \int D\bar{\psi} D\psi D\bar{\phi} D\phi \exp\Big\{ \sum_{k=\vec{k},\omega_n} \sum_{\sigma=\uparrow,\downarrow} \bar{\psi}_{k\sigma}(i\omega - \xi_k)\psi_{k\sigma}$$

$$- \frac{1}{g}\sum_{q=\vec{q},\omega_m} \bar{\phi}_q \phi_q - \frac{1}{\sqrt{\beta V}} \sum_{q k'} \bar{\phi}_q \psi_{-k'+\frac{q}{2}\downarrow} \psi_{k'+\frac{q}{2}\uparrow}$$

$$- \frac{1}{\sqrt{\beta V}} \sum_{q k} \phi_q \bar{\psi}_{k+\frac{q}{2}\uparrow} \bar{\psi}_{-k+\frac{q}{2}\downarrow} \Big\}$$

Note that we can write interaction $\phi - \psi\psi$ term compactly as

$$\frac{1}{\sqrt{\beta V}} \sum_{q k} \left(\bar{\psi}_{k+\frac{q}{2}\uparrow}, \psi_{-k'+\frac{q}{2}\downarrow}\right) \begin{pmatrix} 0 & \phi_q \\ \bar{\phi}_q & 0 \end{pmatrix} \begin{pmatrix} \psi_{k'+\frac{q}{2}\uparrow} \\ \bar{\psi}_{-k+\frac{q}{2}\downarrow} \end{pmatrix}$$

More generally, write

$$\sum_{q k} \phi_q \bar{\psi}_{k+\frac{q}{2}\uparrow} \bar{\psi}_{-k+\frac{q}{2}\downarrow} = \sum_{k k'} \bar{\psi}_{k\uparrow} B_{kk'} \bar{\psi}_{k'\downarrow}$$

where $B_{kk'} = \sum_q \delta_{k+k',q}\, \phi_q$

$$\sum_{q k'} \bar{\phi}_q \psi_{-k'+\frac{q}{2}\downarrow} \psi_{k'+\frac{q}{2}\uparrow} = \sum_{k k'} \psi_{k\downarrow} \bar{B}_{kk'} \psi_{k'\uparrow}$$

and $\bar{B}_{kk'} = \sum_q \delta_{k+k',q}\, \bar{\phi}_q$.

Then $\frac{1}{\sqrt{\beta V}} \sum_{k k'} \left(\bar{\psi}_{k\uparrow}, \psi_{k\downarrow}\right) \begin{pmatrix} 0 & B_{kk'} \\ \bar{B}_{kk'} & 0 \end{pmatrix} \begin{pmatrix} \psi_{k'\uparrow} \\ \bar{\psi}_{k'\downarrow} \end{pmatrix}$

For $\sigma=\uparrow$: $\sum_k \bar{\Psi}_{k\uparrow}(i\omega-\xi_k)\Psi_{k\uparrow}$

$\sigma=\downarrow$: $\sum_k \bar{\Psi}_{k\downarrow}(i\omega-\xi_k)\Psi_{k\downarrow} = \sum_k \Psi_{k\downarrow}(-i\omega+\xi_k)\bar{\Psi}_{k\downarrow}$

$\Rightarrow$ can write

$$\sum_k (\bar{\Psi}_{k\uparrow}, \Psi_{k\downarrow}) \begin{pmatrix} i\omega-\xi_k & 0 \\ 0 & -i\omega+\xi_k \end{pmatrix} \begin{pmatrix} \Psi_{k\uparrow} \\ \bar{\Psi}_{k\downarrow} \end{pmatrix}$$

The action then is written compactly as

$$\sum_{kk'} (\bar{\Psi}_{k\uparrow}, \Psi_{k\downarrow}) \left\{ \begin{pmatrix} i\omega-\xi_k & 0 \\ 0 & -i\omega+\xi_k \end{pmatrix} \delta_{kk'} - \frac{1}{\sqrt{\beta V}} \begin{pmatrix} 0 & B_{kk'} \\ \bar{B}_{kk'} & 0 \end{pmatrix} \right\} \begin{pmatrix} \Psi_{k'\uparrow} \\ \bar{\Psi}_{k'\downarrow} \end{pmatrix}$$

$$Z = \int D\bar{\phi} D\phi \; e^{-\frac{1}{g}\sum_q \bar{\phi}_q \phi_q} \int_{\bar{\chi}\chi} \exp\left\{ \sum_{kk'} \bar{\chi}_k M_{kk'} \chi_{k'} \right\}$$

where $\chi_k = \begin{pmatrix} \Psi_{k\uparrow} \\ \bar{\Psi}_{k\downarrow} \end{pmatrix}$ is Nambu vector of fermion (Grassmann) variables.

General rule $\int D\bar{\chi} D\chi \; e^{\bar{\chi}\hat{M}\chi} = \det(\hat{M}) = \exp(\ln\det\hat{M})$

$$= \exp(\mathrm{tr}\ln\hat{M}).$$

$$\mathrm{tr}\ln\left[(i\omega-\xi_k)\hat{\sigma}_3\delta_{kk'} - \frac{1}{\sqrt{\beta V}}\begin{pmatrix} 0 & B_{kk'} \\ \bar{B}_{kk'} & 0 \end{pmatrix}\right] =$$

$$= \mathrm{tr}\ln\left[(i\omega-\xi_k)\hat{\sigma}_3\delta_{kk'}\right]\cdot\left[1 - \frac{1}{i\omega-\xi_k}\hat{\sigma}_3\delta_{kk'}\cdot\begin{pmatrix} 0 & \frac{B_{kk'}}{\sqrt{\beta V}} \\ \frac{\bar{B}_{kk'}}{\sqrt{\beta V}} & 0 \end{pmatrix}\right] =$$

$$= \mathrm{tr}\ln\left[(i\omega-\xi_k)\sigma_3\delta_{kk'}\right] + \mathrm{tr}\ln\left[\qquad\qquad\right]$$

The first term does not contain field B

$\Rightarrow$ describes non-interacting Fermi gas $\Rightarrow$ omit it.

The 2nd term:

let $$\hat{G}_k = \begin{pmatrix} \frac{1}{i\omega - \xi_k} & 0 \\ 0 & \frac{1}{-i\omega + \xi_k} \end{pmatrix} = G_k \hat{\sigma}_3$$

$$\mathrm{Tr}\ln\left(1 - G_k \begin{pmatrix} 0 & B_{kk'} \\ \bar{B}_{kk'} & 0 \end{pmatrix} \frac{1}{\sqrt{\beta V}}\right) = -\mathrm{Tr}\left\{\sum_{n=1}^{\infty} \frac{(-1)^n}{n}\left[\hat{G}_k \begin{pmatrix} 0 & B \\ \bar{B} & 0 \end{pmatrix}\right]^n\right\}$$

<u>$n=2$</u> is first non-vanishing contribution

$$-\frac{1}{2}\mathrm{Tr}\left\{\hat{G}_k \begin{pmatrix} 0 & B_{kk'} \\ \bar{B}_{kk'} & 0 \end{pmatrix} \hat{G}_{k'} \begin{pmatrix} 0 & B_{k'k} \\ \bar{B}_{k'k} & 0 \end{pmatrix}\right\} = -\frac{1}{2}\sum_{kk'} G_k G_{k'} \cdot \frac{1}{\beta V} \cdot$$

$$\cdot\, \mathrm{tr}\left\{\hat{\sigma}_3 \begin{pmatrix} 0 & B_{kk'} \\ \bar{B}_{kk'} & 0 \end{pmatrix} \hat{\sigma}_3 \begin{pmatrix} 0 & B_{k'k} \\ \bar{B}_{k'k} & 0 \end{pmatrix}\right\} =$$

$$= +\frac{1}{2\beta V}\sum_{kk'} G_k G_{k'} \left(B_{kk'}\bar{B}_{k'k} + \bar{B}_{kk'} B_{k'k}\right)$$

$$\sum_{kk'} G_k G_{k'} B_{kk'} \bar{B}_{k'k} = \sum_{kk'} G_k G_{k'} \sum_{qq'} \delta_{k+k',q}\, \phi_q \cdot$$

$$\cdot\, \delta_{k'+k,q'}\, \bar{\phi}_{q'} = (q' = q,\ k' = q-k)$$

$$= \sum_{kq} G(k)\, G(q-k)\, \phi_q \bar{\phi}_q$$

Clearly the other term gives the same.

Thus $\underline{n=2}$: $+\frac{1}{\beta V}\sum_{kq}\bar{\phi}_q\phi_q\, G(k)\,G(q-k) =$

$$= +\sum_q \bar{\phi}_q\phi_q \frac{1}{\beta}\sum_{\omega_n}\frac{1}{V}\sum_{\vec{k}}\frac{1}{i\omega_n-\xi_k}\,\frac{1}{i\Omega-i\omega_n-\xi_{q-k}}$$

$$= \sum_{q=(\vec{q},\omega_m)} -\bar{\phi}_q(\Omega)\,\phi_q(\Omega)\int\frac{d\vec{k}}{(2\pi)^2}\frac{1}{\beta}\sum_{k}\frac{1}{(i\omega_n-\xi_k)(i\omega_n-i\Omega+\xi_{k-q})}$$

Focus now on the <u>condensate</u> $\phi_{\vec{q}}(\Omega) = \Delta\,\delta_{\vec{q},0}\,\delta_{\omega_m,0}+\phi$

$$\Rightarrow -\bar{\Delta}\Delta\int\frac{d\vec{k}}{(2\pi)^d}\frac{1}{\beta}\sum_{\omega_n}\frac{1}{(i\omega_n-\xi_k)(i\omega_n+\xi_k)} =$$

$$= +|\Delta|^2\int\frac{d\vec{k}}{(2\pi)^d}\frac{1}{\beta}\sum_{\omega_n}\frac{1}{\omega_n^2+\xi_k^2}$$

This means that

$$Z=\int D\bar{\phi}\,D\phi\;\exp-\left\{|\Delta|^2\underbrace{\left(\frac{1}{g}-\frac{1}{\beta}\sum_k\int\frac{d\vec{k}}{(2\pi)^d}\frac{1}{\omega_n^2+\xi_k^2}\right)}_{\text{relative minus sign}}+\dots\right\}$$

Have to look

suggest possible change of sign!

Critical T_c is determined by

$$\frac{1}{g} = T_c\sum_{n=-\infty}^{\infty}\int\frac{d\vec{k}}{(2\pi)^d}\frac{1}{\pi^2T_c^2(2n+1)^2+\xi_k^2}$$

$$\frac{1}{\beta}\sum_{n=-\infty}^{\infty}\frac{1}{\frac{\pi^2}{\beta^2}(2n+1)^2+\xi_k^2} = \frac{1}{\beta}\sum_{n=0}^{\infty}(\ ) + \frac{1}{\beta}\underbrace{\sum_{n=-\infty}^{-1}(\ )}_{\text{Set } n=m-1} = \frac{2}{\beta}\sum_{n=0}^{\infty}(\ ).$$

We know that $\prod_{n=0}^{\infty}\left(1+\frac{4x^2}{\pi^2(2n+1)^2}\right) = \cosh x$ $\qquad \frac{1}{\beta}\sum_{m=0}^{\infty}\frac{1}{\frac{\pi^2}{\beta^2}(2m+1)^2+\xi_k^2}$

$$\Rightarrow \ln(\cosh x) = \sum_{n=0}^{\infty}\ln\left(1+\frac{4x^2}{\pi^2(2n+1)^2}\right)$$

$$\Rightarrow \underbrace{\frac{d}{dx}\ln(\operatorname{ch}x)}_{\operatorname{th}(x)} = \sum_{n=0}^{\infty}\frac{1}{1+\frac{4x^2}{\pi^2(2n+1)^2}}\,\frac{8x}{\pi^2(2n+1)^2} =$$

$$= 8x\left(\sum_{n=0}^{\infty}\frac{1}{\pi^2(2n+1)^2+4x^2}\right) = \operatorname{th}(x)$$

$$\Rightarrow \frac{2}{\beta}\sum_{n=0}^{\infty}\frac{1}{\frac{\pi^2(2n+1)^2}{\beta^2}+\xi_k^2} = 2\beta\sum_{n=0}^{\infty}\frac{1}{\pi^2(2n+1)^2+\beta^2\xi_k^2}$$

$$\Rightarrow 4x^2 = \beta^2\xi_k^2 \Rightarrow x = \frac{\beta\xi_k}{2}$$

$$2\beta\sum_{n=0}^{\infty}\frac{1}{\pi^2(2n+1)^2+\beta^2\xi_k^2} = 2\beta\frac{\operatorname{th}x}{8x} = 2\beta\frac{\operatorname{th}\left(\frac{\beta\xi}{2}\right)}{4\beta\xi} = \frac{\operatorname{th}\frac{\beta\xi}{2}}{2\xi}$$

$$\operatorname{th}\frac{\beta\xi}{2} = \frac{e^{\frac{\beta\xi}{2}}-e^{-\frac{\beta\xi}{2}}}{e^{\frac{\beta\xi}{2}}+e^{-\beta\xi/2}} = \frac{e^{\beta\xi}-1}{e^{\beta\xi}+1} = \frac{e^{\beta\xi}+1-2}{e^{\beta\xi}+1} = 1-2n_F(\xi)$$

where $n_F(\xi) \equiv \frac{1}{e^{\beta\xi}+1}$ is Fermi-Dirac function.

In this way proven that

$$\frac{1}{\beta}\sum_{n=-\infty}^{\infty}\frac{1}{\frac{\pi^2}{\beta^2}(2n+1)^2+\xi_k^2} = \frac{\operatorname{th}\left(\frac{1}{2}\beta\xi_k\right)}{2\xi_k} = \frac{1-2n_F(\xi_k)}{2\xi_k}.$$